30 June

SQL Summary

This section will introduce basic SQL Syntax:

1.1 SELECT Statements:

- SELECT *
- SELECT DISTINCT
- FROM
- As (Aliases)

1.2 Filtering and Sorting

- WHERE Statement
- ORDER BY clause
- SQL COMMENTS

1.3 SQL Operators

- Arithemic operators
- Bitwise operators
- Comparison operators
- Compound operators
- Logical operators

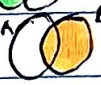
1.4 SQL Aggregate fx (Filtering Grp data)

- SUM()
- AVG()
- MIN() and MAX()
- GROUP BY clause
- HAVING clause

1.5 COMBINING TABLES (Joins & Unions)

- INNER JOIN 

- LEFT JOIN 

- RIGHT JOIN 

- FULL ~~JOINX~~ OUTER JOIN 

- SELF JOIN
- CROSS JOIN
- UNION Operator

1.6 Conditional Logic

- CASE Statements

allows users ... to up of ...

18 June 26

SQL Aggregate functions

- These functions summarize multiple rows into a single value making them essential for analytics, reporting, and trend analysis.
- They are used with :- GROUP BY statements
  - > HAVING statements
  - > ORDER BY statements (mostly in ASC and DSC)

Mostly to segment and to filter results efficiently.

1. COUNT() -> Counts rows
2. SUM() -> Finds totals in numeric columns
3. AVG() -> finds the mean (middle value of numerical column) of numeric values
4. MIN()/ -> Finds the smallest value
5. MAX() -> Finds the largest value

NB: SQL = CRUD

QUERY = Set of instructions given to RDBMS to perform tasks and retrive outcomes

In the SORTING STATEMENT:

>> Simple; here we deal with Assending and Dessending values
>> From 1-10 ASC
>> From 2-9 DSC
>> We use "ORDER BY" statement
>> We also include the conditions :(Becaus, they allow you to Filter data before Sorting it). Without conditions, SQL can Sort all rows. (and we don't want that, do we? No!!) making more meaningful results.
- Also improving performance.

5. MIN-Function: Returns the smallest value in a column.
(With > GROUP BY)

: Lets use the previous table: "Employee's table"
Q: Find the Lowest Salary in each department

Syntax: SELECT MIN(Salary) AS Min_Salary → this will give out a new table name being Min_Sal
FROM employee
~~WHERE department~~
GROUP BY department ID

↳ We are grouping by the department ID because, we want to get the Lowest in each department.

BUT: Excluding the GROUP BY we can get

SELECT MIN(Salary)
AS MIN-salary
FROM employees

(MIN) Lowest Salary

Emp ID	Name	Dept.	Salary
-	-	-	4500

4. MAX-Function: - Returns the Largest/Biggest value in the Column

Syntax: SELECT Max(salary) As Max-Salary
FROM employees

Outcome:

Max-Salary
7000

1. COUNT() Function: Returns the number of rows that match a Condition

The Following table shows a Pizza Orders Table: You are ought to find out how many pizzas were ordered today: Write me a SQL Query Statement to show the outcome.

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Charts, network diagrams, flowcharts, process digrams

let just say its a continuation from the previous transaction or receipts.

PIZZA ORDER TABLE

Order ID	Customer Name	PizzaType
201	Musa	Hawaiian
203	David	Pepperoni
204	Mandla	Margherita
2054	Johnno	Regina
205	Nthaby	Hawaiian

Syntax: SELECT COUNT(DISTINCT Order-id) As number of Orders
FROM Pizza order

Simply means: You count number of id (based on the condition for you to get number of Orders as a new table or outcome from the Pizza table

Outcome:

Number of Orders
5

2. Sum Function: Returns total value from a given Column.

Sales-table

Sale ID	Customer-name	Category	Revenue
1	Rofhi	Electronics	500
2	Thabo	Food	250
3	Mandla	Clothing	350
4	Sipho	Electronics	1000

Wirite a SQL query to calculate the total revenue for each Category.

NB: Always look for the keywords in each question

Syntax: SELECT category
SUM() (revenue) As total_revenue
FROM Sales_table
GROUP BY Category

In this case you Sum up the totals before giving the final answer.

You can further SORT it by ASC or DSC on which category sells more on a given period

Outcome:

Category	Revenue
Electronics	1500
Food	250
Clothing	350

SELECT(*) category
FROM Sales_table
~~WHERE category~~
ORDERS BY ASC / DSC

HAVING STATEMENT: Used to filter groups of data after a GROUP BY operation.

» WHERE: Filters individual rows before grouping
» HAVING: Filters grouped results after grouping.

Sales table

SalesId	Customer	Product	Amount
106	Nthabi	Monitor	800
107	Nthabi	keyboard	500
108	Thando	Speakes	1000
109	Thando	CPU	500
110	Shandu	Laptop	9000

1. Write an SQL query to find customers who spend more than 1000?

First you calculate the sum of each customer, to apply "WHERE" so we can filter the grouped results after grouping our sum totals on each customer.

flowcharts, process ...

Nthab = 1300
Thando = 1500
Shandu = 9000

Syntax SELECT customer_name
SUM(Amount) As total_spent
FROM Sales_table
GROUP BY customers name
HAVING SUM (Count) > 1000

Custom.	Total Spent
Nthab	1300
Thando	1500
Shandu	9000

Simply means: This will return the customer_names with the (Sum Amount giving us a new table AKA - total-Spent; From the Sales-table, where we are going to group the same customers that we want to know the spending amount to know each total value on their receipts. Resulting in a single value of each customer that is greather than > 1000 and also less (if any).

SQL Exercise 3

Student #

SQL CASE Statement

The Product table

SELECT Product_name, Price
CASE
WHEN If Price > 1000 THEN 'Expensive'
WHEN If price BETWEEN 100 AND 1000 THEN 'Mid-range'
WHEN If Price < 100 THEN 'Budget'
END As Price_Category
FROM Products;

1.

Product-id	Product-name	Price	Price-Category
1	Laptop	1200	Expensive
2	Phone	800.00	Mid-range
3	keyboard	45.00	Budget
4	Monitor	300.00	Mid-range
5	Mouse	25.00	Budget

2. The Orders Table

```
SELECT name, amount(
CASE
WHEN Orders ≥ 1000 THEN 'High Value'
WHEN Orders BETWEEN 500 - 999.99 THEN 'Medium value'
WHEN Orders < 500 THEN 'Low Value'
END AS Value-Category
FROM Orders;
```

Customer-name	Amount	Order-value-category
Alice	150.00	Low value
Bob	560.00	~~Low value~~ Medium
Charlie	999.99	~~High value~~ Medium
Diana	45.50	Low value
Ethan	1200	High value

3. The employees Table

```
SELECT name, salary
CASE WHEN 'IT' salary > 80000 THEN 'Senior IT'
WHEN 'HR' Salary > 55000 THEN 'Experienced HR'
ELESE 'Staff'
END AS position-level
FROM employee;
```

Emp-name	Department	Salary	Position_level
John	IT	85000.00	Senior - IT
Sara	HR	60 000.00	~~Staff~~ Experienced HR
Mark	IT	75 000.00	Staff
Lucy	Finance	95000.00	Staff
Tom	HR	55000.00	Staff

4. THES Student Table

Student_name	Score	Grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

```
SELECT name, score
CASE
• ≥ 90 THEN 'A'
WHEN 80-89 THEN 'B'
WHEN 70-79 THEN 'C'
WHEN 60-69 THEN 'D'
WHEN < 60 THEN 'F'
END As Score_grade
FROM Student;
```

5. The deliveries Table

delivery_id	Delivery_time_minutes	Performance
1	45	On time
2	80	Late
3	30	~~Fast~~ Fast
4	65	Late
5	100	Late

```
SELECT delivery_id, delivery_time_minute
>CASE
WHEN ≤ 30 mins THEN 'FAST'
WHEN 31-60mins THEN 'On time'
WHEN > 60min THEN 'Late'
END As delivery_performance
FROM Deliveries;
```

6. THE Tickets Table

Issue_type	Priority	Priority_lable
Long issue	1	Low
Serve down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

```
SELECT issue_type, Priority
CASE
WHEN 3 THEN 'high'
WHEN 2 THEN 'Medium'
WHEN 1 THEN 'LOW'
END As Priority_lable
FROM Tickets;
```

7. The Attendance Table

Student_id	attendance_Percetage	Attentance_Status
1	90%	Excellent
2	60%	Needs improvent
3	96%	Excellent
4	50%	Needs improvent
5	100%	Excellent

SELECT student_id, attentance percetage
CASE
WHEN ≥ 90% THEN 'Excellent'
WHEN 75-89% THEN 'Good'
WHEN < 75% THEN 'Needs improvent'
END As Attendance_status
FROM Attendance table;

8. The Products_Inventory Table

Syntax; SELECT product_id, Stock_qty
CASE
WHEN 0 THEN 'Out of Stock'
WHEN 1-5 THEN 'Low Stock'
WHEN >5 THEN 'In Stock'
END As Stock_status
FROM Products_Inventory

Product_Id	Stock_qty	Stock_Status
1	5	Low-Stock
2	0	Out of Stock
3	25	IN Stock
4	10	IN Stock
5	3	Low Stock

9. The classes Table

Syntax: SELECT subjects, enrolled students
CASE
WHEN ≥ 25 THEN 'Large'
WHEN 10-24 - DA THEN 'Medium
WHEN <10 THEN 'Small'
END AS Class-Size-Category
From classes table;

~~Class-id~~

Subject	enrolled_students	class_category
Math	30	large
English	25	large
Science	15	Medium
Art	5	Small
History	20	Medium

10. The Payments Table

CASE> SELECT amount, Payment method, id
WHEN Payment method = Cash AND AMOUNT ≥ 200 THEN 'Eligible for disco
Otherwise Not 'Eligible'
End as discount-eligibiety
FROM payments

Payment_id	Payment_method	Amount	Discount_eligibility
1	card	50.00	Not eligible
2	cash	200.00	Eligible for discount
3	card	150.00	Not eligible
4	PayPal	75.00	Not eligible
5	Cash	300.00	Eligible for discount

Platforms and tools for Data Analytics!

→ learning and Productivity Platforms-

1. **Draw.io** (tool) → Database Diagrams : Web-based diagraming tool used for creating flowcharts, process diagrams, organizational charts, network diagrams, UML diagrams, and other visual representations of Informations.

2. **Google Colab**: Cloud based platform that allows you to write, run, and share python code directly in your Web browser with access to computing resources including CPU's, GPU's and TPU's.

3. **Google looker Studio**: (data Studio) is a suite of tools and platforms launched by google, each serving a different purpose depending on the users role and needs. Its goal is to simplify tasks related to artificial intelligence, automation and data Analysis, making these technologies accessible to both technical and non-technical users.

4. **Miro**: A visual workspace where teams can design, brainstorm and plan any project and visualize it in real time.

5. **Lovable**: serves as a comprehensive AI-powered data (analytics) and visualization tool, ideal for teams to translate complex user into actionable insights.

6. **Google Sheets**: Spreadsheet Application that allows users to create, edit, and collaborate on spreadsheets online. It is part of Google Space and functions similarly to Microsoft excel But with Added Advantage of Cloud-based Collaboration. It works in any Web browser, doesn't require any instalation, making it accessible on computer, tablets and smartphones

7. Microsoft Excel: Powerful spreadsheet software used to Organize analyze and visualize data for business and personal purposes. It also allows users to create workbooks consisting of multiple worksheets which are made up of rows and Columns

8. Canva: Online graphic design tool that allows users to Create professional-quality designs easily using drag and drop tools used for, presentations, posters, flyers, logos and videos.

9. data breks: Is an unified data and AI plat-form that runs both anylitical and operational workloads on one open governed foundation. It ~~one~~ enables Organizations to build BI and AI applications, stream and analyse data and process daily operations to consistance Security, performance and governance. without adding complexity or having to Copy data.

Terms and meanings

1. BI: BUsiness Intelligence
2. AI Artificial intelligence
3. KPI: key Performance Indicator
4. ML: Machine learning
5.

- they help track progress and, data drive. driven decisions to give you clear picture of current progress and future su

7 Modules *

Data Analytics 01

15 May. 26

What is data Analysis(t)?

Data: > Collection of raw, unorganized facts (Numbers, text or observations. Only becomes meaningful information Only when analysed, contextualized and processed,

Data Analyst: A person (group /team) Responsible for collecting, and analyzing data or raw information to make informed or meaningful for business decisions)
This team or person uses various softwares Like
1. Excel
2. SQL or Python
they identify trends, patterns and correlations. Working with raw data from previous sources, ensuring its accuracy and completeness before applying "Stastical METHODS and Analytical TOOLS to extract valuable insights.

They focus more visualizing data through (Charts dashboards and Reports) Collaborating across departm to address business challanges and improve performance

① Statistical Methods and Tools

(i) Excel: Versatile Spreadsheet software developed by microsoft, used for ORGANIZING data, PERFORMING Calculation and Analysing information.

(ii) Python: is a CORE Programming language used to tranform raw data into business insights: Mainly Used to
- TREND & PARTTERN IDENTIFICATION: Used *Long Side to Excel and SQL to identify correlations and trends within raw data Sources.

• Performance Analysis: A key tool for finding performance drivers and solving complex busines challenges.

• Also in Advanced Analystics, can be used as primary tool (Language for Junior Data Scientists performing, Predicting analytists and machine learing.

(iii) Structured Query Language: used to store, retrieve and manipulate data within relational database. Its PRIMARY GOALS:

• Extraction data & Retrieval: Executing queries to pull a specific data from one or multiple tables. Using "SELECT" Statements, filters ("WHERE") and Joins.

• Data cleaning & PREPARATION: Ensuring data accuracy and Completness before anglysis by fixing errors and handing missing values,

• Data Manupilation: Summerizing and aggregating data (using Sum. Avg. Count.) to identify trends, partiens and Correlation.

• Data Management: Creating and Modifying the Structure of database and tables as well as managing records through inserts, update and delete Operations.

• Performance Optimization: Using Index and Stored Procedures to ensure fast quering and reusable logic for complex analytical task

Key Responsibilities of Data Analyst.

1. Collect and clean data: Gather data from different sources and fix errors, missing values, duplicates, and formatting issues so its analysis-ready.

(Gather/collect data)

- Web Scraping - Uses tools like BeatifulSoup or Octoparse to extract data from websites.
- Surveys and Forms - Use platforms like ~~Survey~~ Survey monkey or Google forms for direct feedback.
- API's & Databases : Pull programatic data direct from official databases.

(2) Clean Data / Data Wrangling Process: Transforms raw data inputs into coherent data by addressing common quality issues;

- Remove Duplicates : Scans your data base and remove exact duplicates rows to prevent skewed analysis.
- Handle Missing Values : Identify blank or null fields Desides wether to drop the incomplete rows or fill them using average or estimates.
- Standardize Formats : Ensures data formats, Phone # and currencies follow a consistent rule, eg (Changing "12/10/2026" and "10-12-26" to YYYY-MM-DD)
- Fix Typos & Structural Errors : Trim extra blank spaces fix misspellings, and merge inconsistent Categories (e.g merging "Sandton," Sandton, SA, Saidton)

2. Analyze data: Use Excel, SQL, Python (plus basic statiscs) to find trends, patterns and performance drivers.

3. Visualize Insights: Build charts, dashboards, and visualize that make complex results easy to understand

4. Report & Present: Turn finolings into clear reports and slides with key insigths and practical recommendation

5. Solve Bussiness Problems: Translate bussiness queries into analysis and propose actionable improvements based on evidence.

6. Collaborate w Stakeholdes: Work w marketing, finance ops to understand needs, define metrics deliver insights

7. Mantain & Monitors Reporting: keep data pipelines dashboard, and recurring reports accurate, updated and consistent

8. Drive Decisions w data: Provide objective evidence that powers smarter, faster business decisions.

Tools and technologies (* Industrial toolkit*)

FOR: • Data cleaning:
(i) Microsoft Excel
(ii) Google Sheets
(iii) Structured Query Language

FOR: Data Visualization:
- Microsoft excel
- Google Sheets
- Power BI

FOR: Data Presentation

- Microsoft Powerpoint
- Google Slides
- Canva

FOR: Project Management

Miro Canva . Figma
Microsoft Excel
Google Sheets

FOR: Databases:

Databricks , Snowflake
Microsoft SQL server
My SQL Workbench

FOR Version Control

GitHub
Git CLI & branching
Pull requests & merges

June Intake
SQL: Structured Query Language: 28050020402